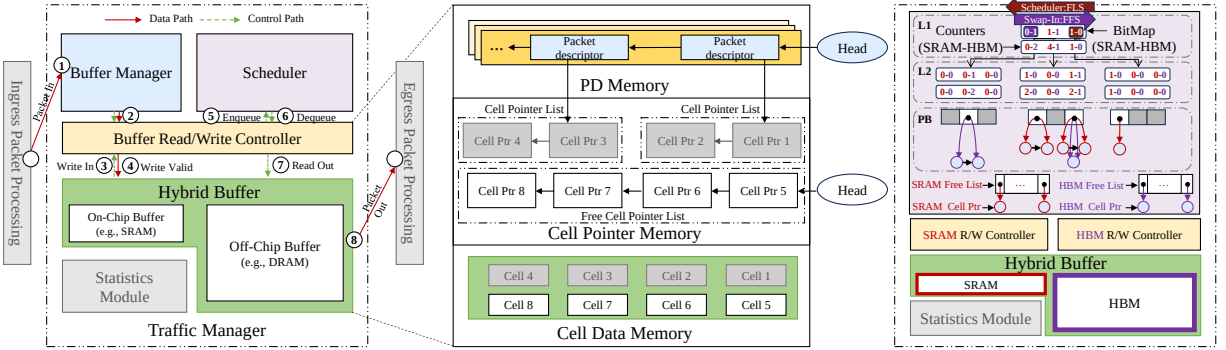

Data Path
Control Path
Ingress Packet Processing
Packet In
Buffer Manager
Scheduler
Enqueue
Dequeue
Buffer Read/Write Controller
Write In
Write Valid
Read Out
Hybrid Buffer
On-Chip Buffer
(e.g., SRAM)
Off-Chip Buffer
(e.g., DRAM)
Statistics Module
Traffic Manager
Packet Out
Egress Packet Processing
(a) Traffic Manager
Packet descriptor
Packet descriptor
Head
PD Memory
Cell Pointer List
Cell Pointer List
Cell Ptr 4
Cell Ptr 3
Cell Ptr 2
Cell Ptr 1
Cell Ptr 8
Cell Ptr 7
Cell Ptr 6
Cell Ptr 5
Head
Free Cell Pointer List
Cell Pointer Memory
Cell 4
Cell 3
Cell 2
Cell 1
Cell 8
Cell 7
Cell 6
Cell 5
Cell Data Memory
(b) Buffer Architecture
Scheduler:FLS
Swap-In:FFS
L1
Counters
(SRAM-HBM)
BitMap
(SRAM-HBM)
L2
PB
SRAM Free List
HBM Free List
SRAM Cell Ptr
HBM Cell Ptr
SRAM R/W Controller
HBM R/W Controller
Hybrid Buffer
SRAM
HBM
Statistics Module
(c) Themis